



Case Study Project: Final Report

A SNAPSHOT OF U.S.A. ENERGY PRODUCTION
WITH ADVANCED VISUALISATION

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Executive Summary

This project visualises the development of renewable energy technologies in the United States from 2013 to 2019, using data from the World Resources Institute's Global Power Plant Database (2024). The visualisations focus on two key indicators: (1) total annual electricity generation (GWh) from all energy facilities and (2) generation by fuel type. The aim is to assess the uptake of renewable energy in the context of global commitments to achieve net zero emissions by 2050.

Developed in RStudio, the project employs the MapDeck package to efficiently render large spatial datasets using GPU acceleration through Deck.gl. Integration with Shiny enables an interactive dashboard where users can explore national and temporal energy trends through dynamic filters and sliders. The primary visualisation techniques employed include an interactive browser-based geographical map, detailed 3D projections at a granular level, and a supporting line chart subplot.

The analysis reveals that fossil fuels, particularly coal and gas, continue to dominate energy generation, while renewable sources such as solar and wind show gradual, steady growth. This suggests that although diversification is occurring, dependence on non-renewable sources remains substantial.

Key visualisation challenges include the limited use of colour, which currently only differentiates fuel types rather than representing percentage changes in energy output, reducing the ability to perceive subtle temporal variations. Additionally, the line plots make cross-comparisons difficult due to scale distortions that affect the perceived steepness of growth or decline. These insights emphasise the importance of effective colour encoding, scale, and visual clarity in accurately communicating data trends and avoiding misleading interpretations.

Overall, this project demonstrates the potential of combining interactive geospatial visualisations with temporal analytics to track the adoption of renewable energy. By highlighting both the progress and ongoing reliance on fossil fuels, it serves as an informative and practical analytical tool for readers seeking insight into U.S. progress toward achieving net zero emissions by 2050.

1. Introduction

The visualisations in this proposal aim to communicate the development of renewable energy technologies in the United States of America (USA) through the years of 2013 to 2019. The primary quantitative indicators to track meaningful progress are:

1. Annual electricity generation (GWh) from all energy generating facilities, and
2. Annual electricity generation by fuel type.

The visualisations in this proposal aim to communicate the development of renewable energy technologies, in comparison with other non-renewable energy types, in USA through the years of 2013 to 2019.

The main data visualised is collated from a publicly available source: the World Resource Institute Global Power Plant Database (Global Power Plant Database - Datasets, 2024). The initial plan to capture the national trends, a separate dataset was sourced for this purpose. However, the U.S. energy production data extracted from the U.S. Energy Information Administration (U.S. Energy Information Administration, 2024) did not capture statistics from diverse fuel types like the main Global Power Plant Database. Therefore, to make insightful inferences about national trends, only one consolidated database was required for this visualisation task.

The purpose of these visualisations is to assess the uptake in renewables in light of a global commitment to achieving net zero greenhouse gas emissions by 2050. The main motivation behind this case study is to build informative and digestible graphics to educate readers on the U.S. efforts in meeting these commitments. To achieve this, the main visualisation utilises animations to emphasise the variations in renewable energy consumption over time. This gives a summarised snapshot of if the general trend within the short time frame. The supporting visualisation empowers readers to understand the trends at a granular level. That is, slicing and examining the trends by year or fuel type, at a self-directed pace.

The visualisation was developed in R-Studio utilizing the MapDeck package to visualize spatial data. MapDeck was selected as the best package for this task as it is built to plot large datasets using Deck.gl as the core library that provides GPU acceleration. This makes rendering the 3D spatial map much faster than doing so on the CPU. It is also compatible with the Shiny library, an app.R file with UI and server code. It provides an interactive interface for the user to directly explore the data with sliders or filters for deeper analysis.

2. Results and Outputs

2.1 Background research and investigations

During the preliminary research for a suitable dataset for this visualisation task, the Global Power Plant Database was shortlisted as the best dataset due to its visualisation power. The database covered approximately 30,000 power plants from 164 countries and contained each entry contained information on plant capacity, generation and fuel type. I first found this dataset on Resource Watch which reformatted the dataset and displayed the visualisation on their website as a scatterplot of facilities on a map utilising Mapbox from MapDeck. This sparked the inspiration on the approach for complex and dynamic spatial visualisation with MapDeck.

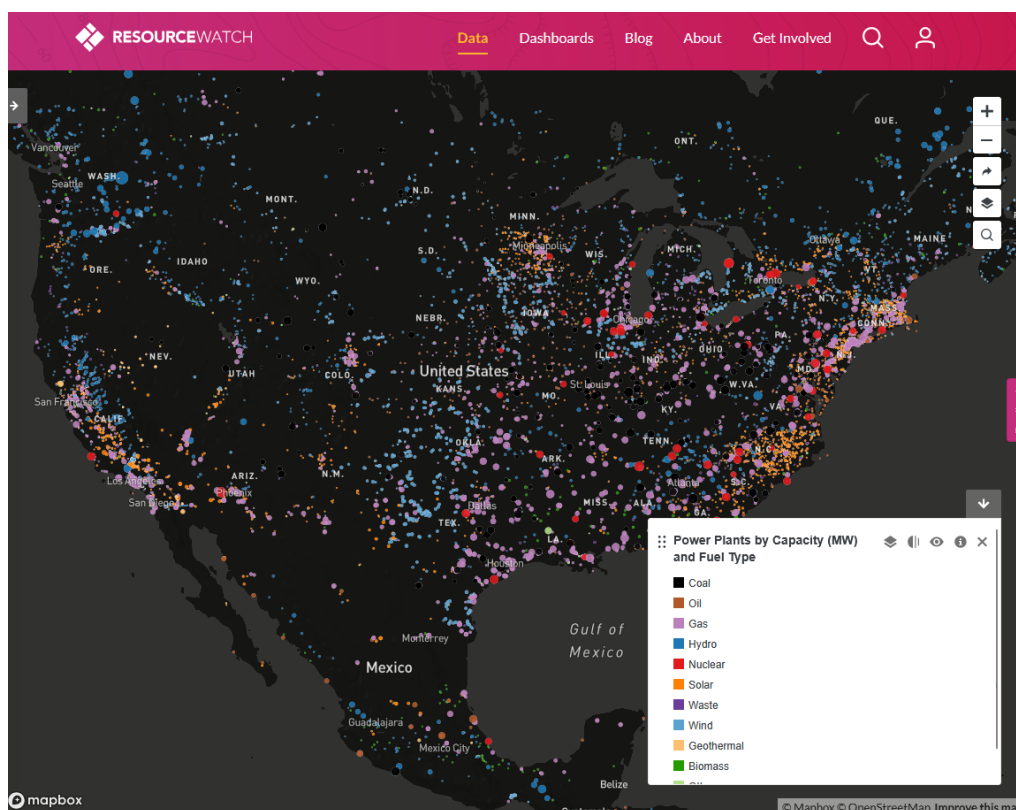


Figure 1: Resource Watch – Global Power Plant Database Visualisation

Further research on the 3D mapping possibilities led me to a medium article “3D mapping global population density: How I built it” (Mapbox, 2018). The writer documented his approach in successfully developing a visualization of global 3D projections at a granular level, in an interactive, browser-based map. He noted that few had attempted creating a visualization at this scope, mostly because the technology (i.e., Mapbox) had not yet existed.

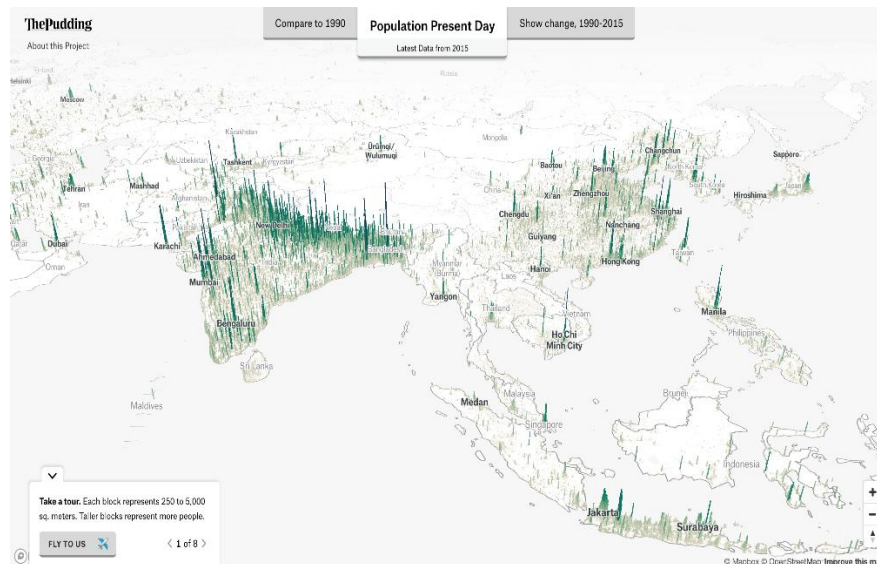


Figure 2: 3D mapping utilising Mapbox

However, although the visualisation proved that 3D projections are possible with Mapbox, it was developed entirely in front-end code using Javascript and Mapbox Studio to create tile layers for web maps. This was a conflicting approach to my initial plan, using the MapDeck package in R environment to build similar visualisations. Hence, an article found on github.io (*Tips & Tricks*, 2020) with tips and tricks provided useful code on how to dynamically add and remove layers with a Shiny app. This made it possible to create a 3D mapping on Mapbox with interactive features for users exclusively on R.

2.2 Data massaging and manipulations

Since the objective of the visualisation is to explore national energy trends in the USA from 2013 to 2019, only USA energy statistics were extracted from the Global Powerplant Database. The USA data included regions like Puerto Rico, the U.S. Virgin Islands and Alaska. By including these points, the viewer is unable to capture the nuance change in yearly national energy trends with a zoomed-out map. To keep the visualisation focused and easy to interpret, the analysis centres on the contiguous U.S., providing a clearer view of national energy trends without the distortion caused by distant territories.

Locations of all the energy powerplants in the U.S.

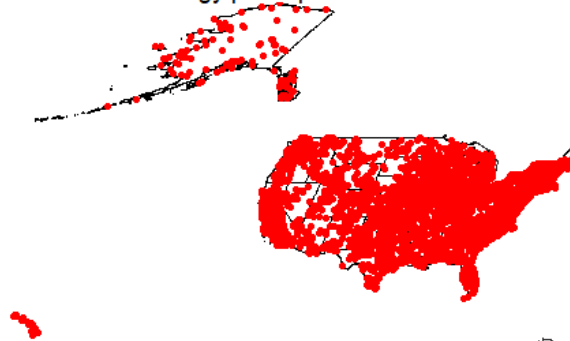


Figure 3: Locations of all the energy facilities in the U.S. including distant territories

Some powerplant facilities were removed as there were missing values in the 2018 and 2019 energy generated statistics, there were 883 and 174 missing values respectively. Although there is some data leakage here, it still does not compromise the effectiveness of the visualisation. Instead, removing incomplete records helps maintain data quality and ensures that trends are meaningfully captured. Nonetheless, this approach acknowledges the potential risk of misrepresenting trends by slightly underestimating overall values.

The data was cleaned to retain only important information, like year, fuel-type, total energy generated, in both long and wide formats. To prepare the data for mapping, a regular grid of square cells was generated using a projected coordinate reference system (CRS). This provides a standardised framework for aggregating powerplant data across all years. If a powerplant falls within that grid cell, their energy generation data will be grouped spatially with other powerplants located in that cell. This allows for an aggregation of energy generation of each fuel type in that contributed by powerplants within each cell. For simplicity, the dominant fuel type (highest generated value) is assigned to each cell. This allows each grid cell on the map to be coloured according to its leading energy source.

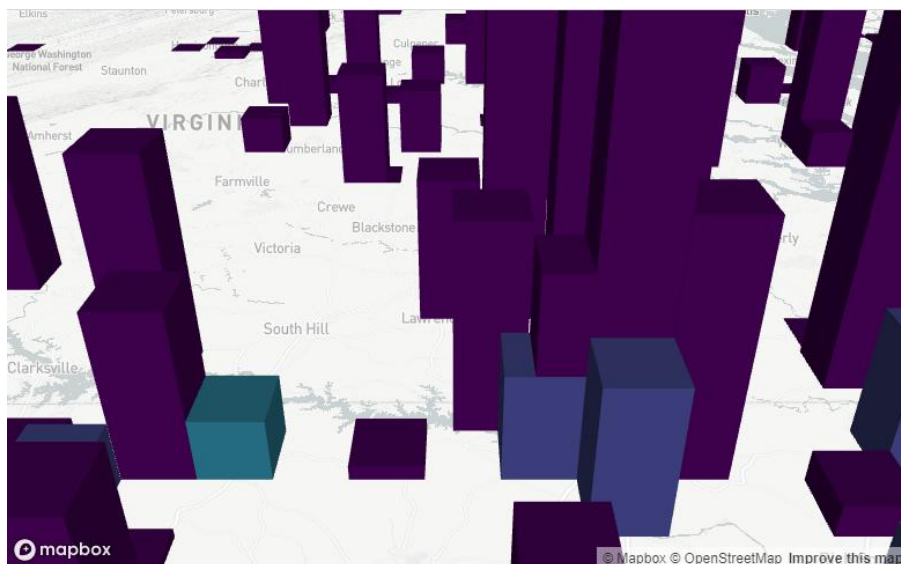


Figure 4: High-density of facilities in some areas cause overlaps in fuel types

2.3 Code description implemented

The main implementation of the interactive mapping and trend integrates Mapdeck for geographic representation and ggplot2 for trend analysis. Both plots are dynamically linked by a single year slider coded in Shiny app.R. The main user interface consists of a sidebar and a main display. The sidebar includes a single year slider that acts as a filter by year range starting from 2013, and doubles as an animation feature. The fuel source dropdown selection box, allowing filtering of specific energy types or viewing all fuels collectively. There is a legend checkbox to show or hide the colour assigned fuel types.

The server function synchronizes both visual components, map and line chart, instantaneously through a reactive/observe function which re-draws the map with every

call. The updates are triggered by user inputs from the user interface side panel. The map is rendered with polygons representing precomputed grid cells. Each cell is colour coded by dominant fuel type and the 3D effect is created by elevating the cell by total generated gigawatt-hours (GWh). The line chart uses ggplot2 and visualises the cumulative trends starting from 2013 up to the selected year. The line chart is created from a separate wide format dataset from the precomputed map rendering dataset.

The main animation comes from the year slider that loops through all the years at an interval of 1000 for the forward temporal walkthrough of the dataset. Both visualisations respond simultaneously, maintaining time consistency for a cohesive data representation and trend analysis.

2.4 Produced visualisation outputs

USA Power Plant Energy Production in Gigawatt-Hours (2013–2019)

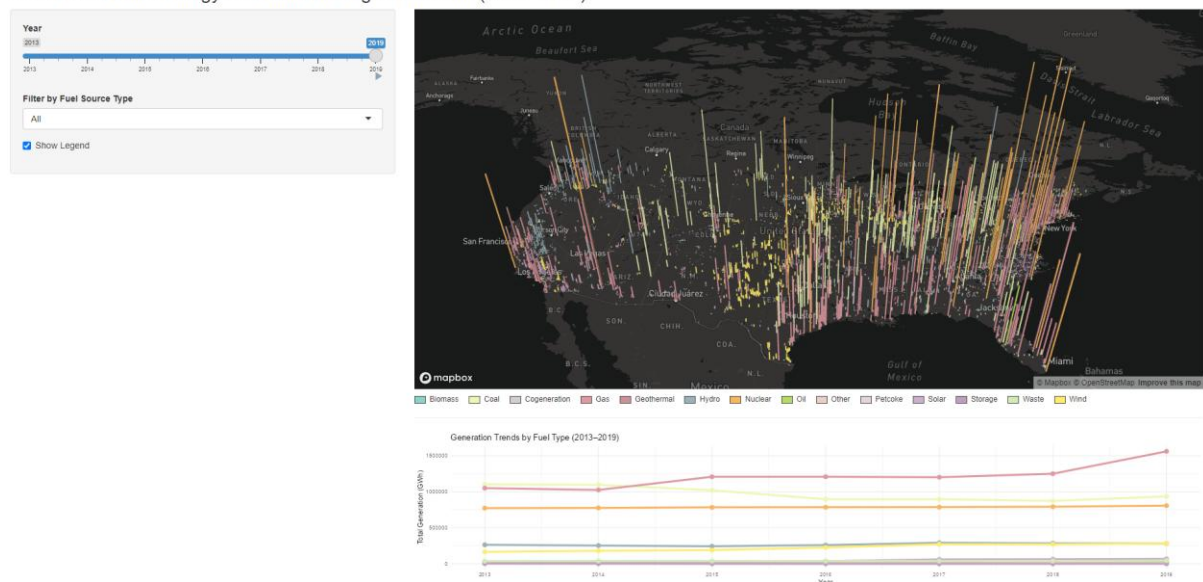


Figure 5: Full visualisation output

USA Power Plant Energy Production in Gigawatt-Hours (2013–2019)

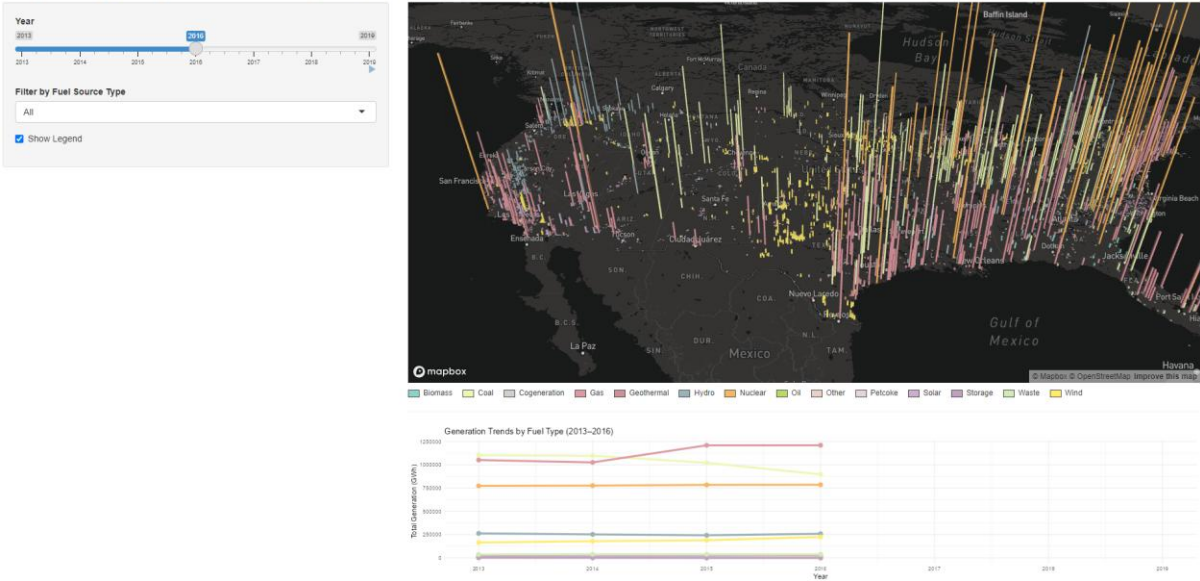


Figure 6: Full visualisation output filtered by year

USA Power Plant Energy Production in Gigawatt-Hours (2013–2019)

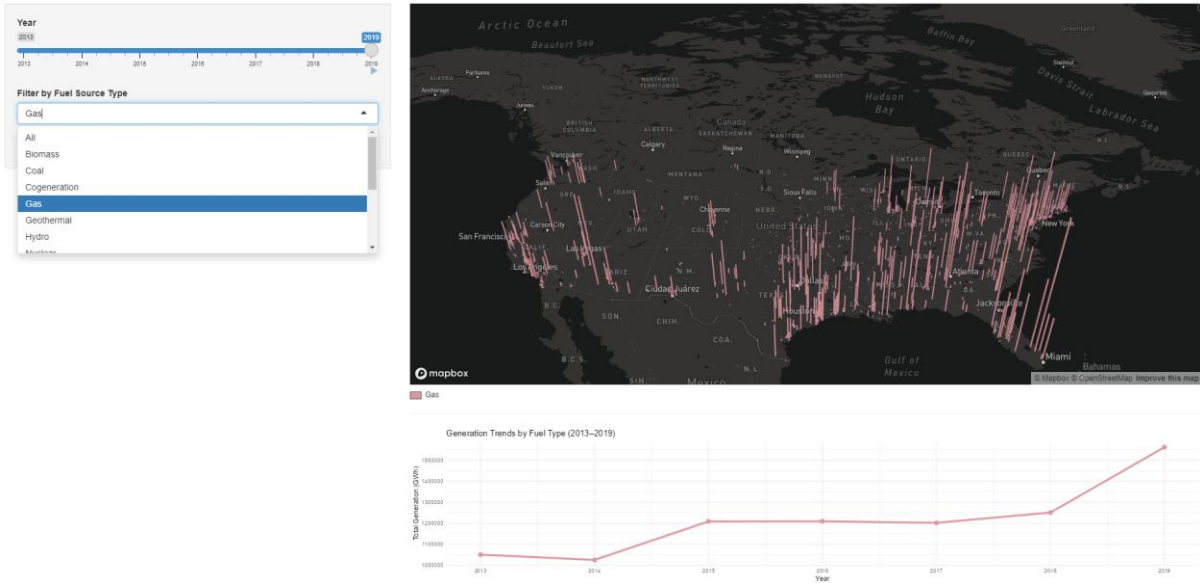


Figure 7: Dynamic changes in visualisation with filter by Fuel Source Type

USA Power Plant Energy Production in Gigawatt-Hours (2013–2019)

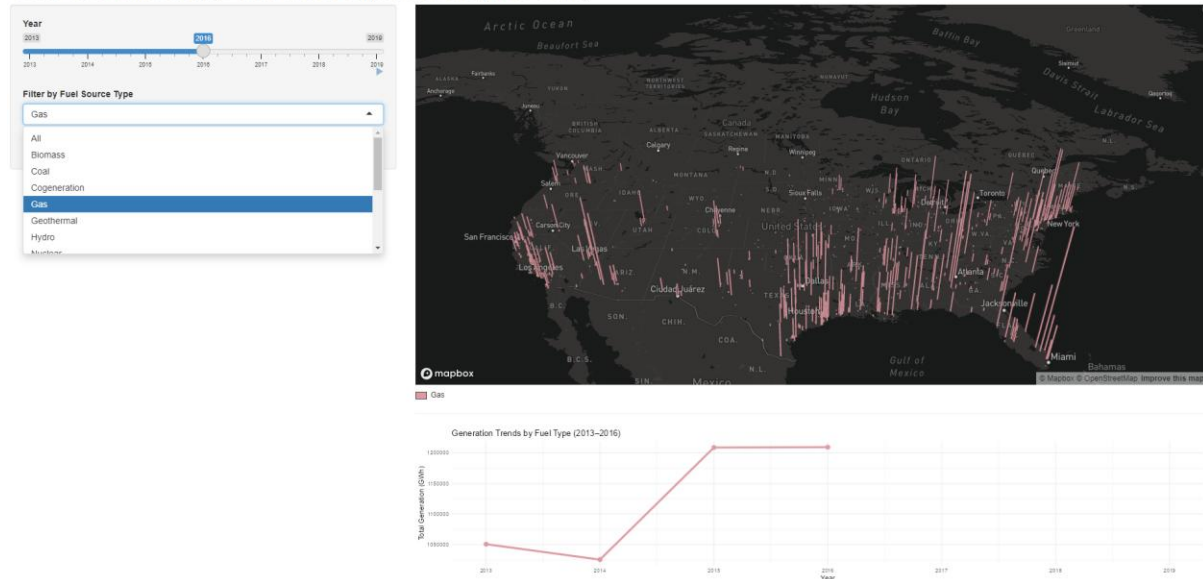


Figure 8: Dynamic changes in visualisation with filter by Year

2.5 Effective visualisation issues

A key issue with the current visualisation is the ineffective use of colour to represent percentage changes in energy generation over the years. Current application of colour is used solely to categorise cells by dominant fuel type. This categorical scheme limits the ability to incorporate an additional colour dimension that could convey granular changes in energy output. As a result, subtle increases or decreases in energy generation across cells are not easily perceived. Furthermore, the map's wide scale and the narrowness of individual cells make it difficult to distinguish temporal trends, causing important patterns of change to be visually lost.

Another effective visualisation issue is that in the line plot for the full output, it is difficult to make meaningful comparisons both between fuel source types and within each individual fuel source. The slopes do not accurately capture the trends because they are relative to the total generation of other fuel sources. This violates the 45-degree rule, as the differences in scale and aspect ratio distort the perceived steepness of the slopes, making growth or decline appear less significant than they actually are.

2.5 Insights and knowledge gained

The analysis highlights how different fuel sources have contributed unevenly to total energy generation over time. Fossil fuels, particularly coal and gas, continue to dominate the energy mix, maintaining relatively stable generation levels with minor fluctuations. Renewable sources such as solar and wind show gradual but consistent growth, reflecting a slow transition towards cleaner energy. These patterns suggest that while diversification of energy sources is underway, large-scale dependence on non-renewable fuels remains a predominant feature of the energy landscape.

From a visualisation perspective, the process underscored the importance of effective visualisation theory to accurately communicate trends. Based on the visualisation issues identified, it demonstrates how visual design can lead to misleading interpretations. This knowledge can inform future visualisations to more effectively convey changes in energy generation across time and categories.

3. Effectiveness of Visualisation Outputs

3.1 Use of negative space

The 3D charts on the map stand out and are enhanced by the dark negative background space with faint white lines used to distinguish physical and political geography. The negative space emphasises the granular details when comparing the height and clusters of the bar charts from 2013 to 2019. This visual balance helps draw attention to key variations in energy generation without overwhelming the viewer. By minimising background distractions, the use of negative space improves depth perception and supports clearer comparison across years and fuel types.

3.2 High Data to Ink Ratio

Every visual element in the visualization maps directly to an important feature of the data, making it highly effective for communication. Brief annotations as a tooltip feature on the map convey key information and help readers interpret the results, when required. The line chart maintains a strong data-to-ink ratio by focusing on trends without unnecessary decorative elements, ensuring that attention remains on the data itself.

3.3 High audience engagement

Features such as tooltips, zoomable map elements, and animations allow viewers to explore temporal trends, through an interactive and intuitive dashboard. The combination of 3D map elements and time-based comparisons across years sustains user interest by revealing patterns that unfold through interaction. This interactivity promotes active rather than passive viewing to improve data comprehension overall. Hence, encouraging users to draw their own insights, resulting in a more meaningful and memorable engagement with the data.

4. Update Tasks and Project Timeline

MXB362: Case Study Project

Project start: 8/18/2025
Display week: 1

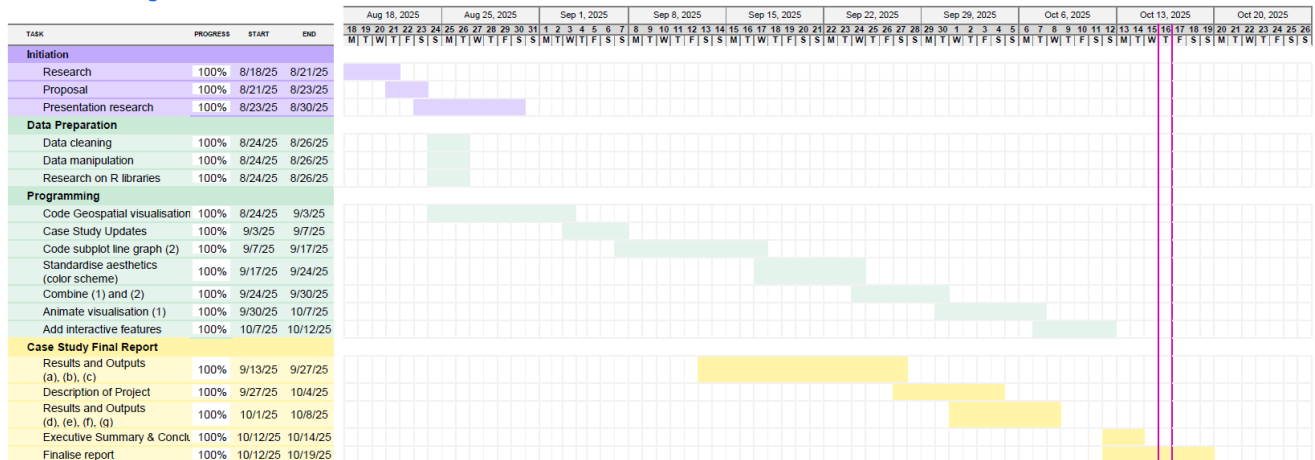


Figure 9: Updated Project Timeline Gantt chart

5. Problems

A major challenge is efficiently precomputing the cell grids and aggregated results. Each year contains approximately 7,000 entries to process, and after precomputation, there are roughly 5,000 entries used for visualization. This stage is computationally expensive but necessary to ensure smooth animation transitions and filtering. To address this, I attempted to convert the precomputed data to JSON format so it could be saved and reused for future visualizations. This approach makes loading the visualization more efficient, as it avoids repeating the precomputation if the data is not already cached in R. However, due to the complexity of the precomputed data structure, JSON proved insufficient. The precomputed data is stored as a list, with each year from 2013 to 2019 represented, and each cell aggregates statistics from multiple power plants, including coordinate geometry stored as a list. This structural complexity makes it difficult to significantly reduce precomputation time simply by saving and reloading the data from an external file. Hence, it would still take approximately 15 mins to finish the precomputation stage of the data preparation process.

A potential issue arises from this data preparation approach when mapping by dominant fuel type within each cell. This challenge is particularly evident in regions with a high density of power plants, such as the eastern and western U.S., where multiple fuel types may overlap within a single cell. Consequently, some fine-grained patterns may be obscured, and the aggregated data may slightly misrepresent the exact mix of fuels in these dense areas. However, the dominant fuel type and overall generation trends are still accurately captured, ensuring that meaningful insights can be drawn at the national scale. This highlights the trade-off between spatial accuracy and aggregation for efficient visualization.

6. Project Journal

Week 1: Research for visualisation proposal

Dates: 13, 14, 15 August

Total Hours: 8 hours

Summary of work completed: Time went into research for what possible visualisations can be done on Matlab and R. Initially I was exploring 3D visualisations on maps like stacked charts on geolocation but found that it may not be the best course of action. I decided to prioritise animations to capture the time series aspect instead of a static chart. Through my research, I found that RStudio has the best visualisation packages that communicate the desired outcome effectively.

Week 9: Initial development of visualisation 1

Dates: 19, 20, 21 September

Total Hours: 20 hours

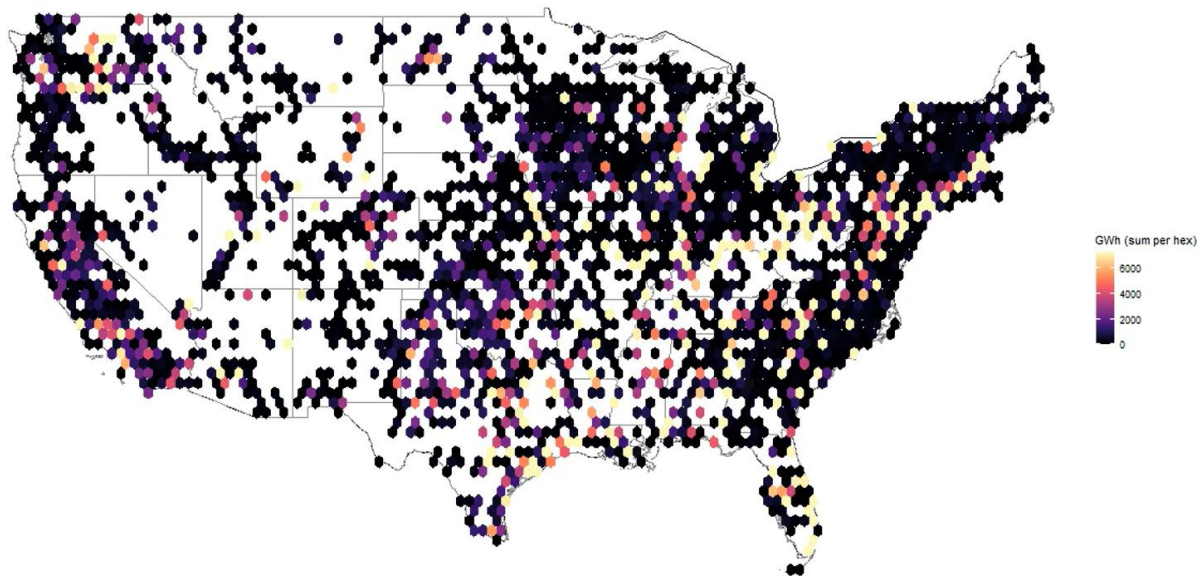
Summary of work completed: Time went into research for R packages to produce map-based visualisations. The initial plan was to create an informative visualisation for Australia. However, in loading the data, I found that there were too many missing values in the electricity generated with little estimated electricity generated values to replace the missing values. So, I've decided to use powerplant data from the U.S. which has more datapoints to create a more complex visualisation. Missing values from 2018-2019 were removed, this prevents loss in data points for the last 2 years which may bring up questions to the viewer. An initial visualisation is created with all the locations of the powerplants to gain an understanding of the data before jumping into animating it. Some challenges like too many data points clustered together, thus making it incomprehensible, needs to be addressed to improve visualization communication. Too many clustered data points do not aid in readers understanding of underlying trends and structures.

Week 9: Exploring MapDeck visualisation

Dates: 25, 26 September

Total Hours: 10 hours

Summary of work completed: Research into applications of MapDeck and the possibility of 3D charts on map visualisation. The plan is to move from plotly, ggplots2, hexbin and gganimate which limit the possibility of developing complex, multidimensional maps. I tried playing around with hexbins at first to visualise the changes in energy generated at a national level, without the dimension of fuel type yet. I found that the 2D representation did not effectively convey the temporal changes. The visual appeared cluttered and disorganised, making it difficult for viewers to distinguish changes across different years.



Mapdeck was found to be the best approach that allows combining multiple data dimensions into a single, interactive visualization. I've also searched for how shiny app.R can be integrated into Mapdeck for a dashboard look and user-controlled functions.

Week 10: MapDeck visualisation implementation

Dates: 29, 30, 5 October

Total Hours: 20 hours

Summary of work completed: First, an account was created with MapDeck. A free tier account was created to use MapBox, which requires a public token key to even use the environment. I tried playing around with the functionality with just 2019 data. I've found that the best function for my 3D chart data is `add_grid`. Afterwhich, it was an iterative process to figure out the best cell size for the grid, the elevation scale for a 3D feature, and assigning colour to each fuel type. As well as, how to ensure the map focuses on U.S. territories and is not zoomed out to the rest of the world. After completing the preliminary visualisation to understand the MapDeck environment, I needed to build a grid foundation for mapping. This step was applied to data in 2019 first. At this stage, after creating my own grid, I needed to use a different MapDeck function which was `add_polygon`.

Week 11: Shiny app.R integration with MapDeck

Dates: 7, 10 October

Total Hours: 15 hours

To scale up from just one year data, I need to test building my visualisation for 2013-2019. I tried implementing Shiny app.R using the resource I found from GitHub.io. The precomp function is created here where it iteratively calls the data preparation step created last week for data in each year. The Shiny UI and Server are developed first before adding, testing and changing the interactive features until they work cohesively with the map. Testing involved ensuring that all reactivity worked seamlessly and efficiently across years. Adjustments were made to optimise loading time and visual consistency. Before the precomp function was developed, I made the Shiny app load each year just as when

it was called. It meant that loading each year's data was slow and at times seemed unresponsive. Hence, the precomp function was developed to ensure the computationally expensive tasks were completed first so that dynamic updates to the map were done fast and instantaneously.

Week 12: Line Chart subplot implementation and uniform UI update features

Dates: 16, 17, 18 October

Total Hours: 25 hours

The line chart subplot data preparation function is created first by converting the powerplant data into a wide format suitable for Shiny app.R visualisations. The line chart was created using ggplot2 as it can also be integrated into Shiny. It took a lot of time synchronizing the UI and server logic so that changes to the year slider, fuel type filter, and legend display would dynamically update both the MapDeck visualization and the line chart simultaneously. Integrating the spatial and tabular components of the data in a reactive framework was complex. Debugging reactive dependencies and maintaining consistent updates between multiple outputs (maps and plots) required careful design and iterative refinement. After which, the visualisation produced was the final case study output presented in the report.

7. Conclusions

The main goal of the visualisation is to compare and analyse the energy generated in gigawatt-hours (GWh) of different fuel types (between renewable and non-renewable energy sources) across the years 2013 to 2019. It also allows the viewer to locate the powerplant on the map, to determine the concentration of facilities. The subplot line chart provides additional information on the total yearly trends of the production of individual energy sources. The goal of the visualisation is also to reveal the magnitude of the temporal trends in energy generated by fuel type.

Despite challenges such as data leakage and long computation times, the visualisation successfully achieved its objective of enabling clear, interactive exploration of multi-year energy trends. Future improvements could focus on optimising performance and making more effective use of colour to show yearly percentage changes. One example is incorporating a stacked bar chart that highlights growth or decline from the previous year through green and red tones.

Overall, the final project output demonstrates a clear understanding of how spatial and temporal data can be integrated into an interactive dashboard. To support meaningful analysis of energy generation trends over time.

7. References

Global Power Plant Database - Datasets. (2024). Wri.org.

<https://datasets.wri.org/datasets/global-power-plant-database>

Tips & Tricks. (2020). Github.io. https://symbolixau.github.io/mapdeck/articles/tips_tricks.html

Mapbox. (2018, October 29). *3D mapping global population density: How I built it*. Medium; maps for developers. <https://blog.mapbox.com/3d-mapping-global-population-density-how-i-built-it-141785c91107>

8. Appendices

8.1 Precomp() Function

```
# --- PARAMETERS ---
cellsize <- 10000      # Size of each grid cell in meters (10 km x 10 km)
elev_scale <- 50       # (Unused in this snippet, could be used for vertical scaling in 3D maps)
years <- as.character(2013:2019) # Vector of years to iterate over

# --- PREP: Create a common grid extent so grids align across years ---
# Convert input data to sf (if not already)
pts_all <- if (!inherits(pp_long, "sf")) st_as_sf(pp_long, wkt = "geometry", crs = 4326) else pp_long

# Transform points to metric projection (EPSG:3857) for consistent distance-based grids
pts_all_m <- st_transform(pts_all, 3857)

# Create bounding box around all points
bb <- st_as_sfc(st_bbox(pts_all_m))

# Create a square grid covering the bounding box
grid_all_sfc <- st_make_grid(bb, cellsize = cellsize, square = TRUE)

# Convert grid to sf object with unique cell IDs
grid_all <- st_sf(cell_id = seq_along(grid_all_sfc), geometry = st_sfc(grid_all_sfc))

# --- SET COLOUR PALETTE FOR ALL FUELS ---
all_fuels <- sort(unique(pp_long$primary_fuel))
pal_base <- "set3"
n_fuels <- length(all_fuels)
maxpal <- RColorBrewer::brewer.pal.info[pal_base, "maxcolors"]

# Generate color vector for all fuel types
cols_vec <- if (n_fuels <= maxpal) RColorBrewer::brewer.pal(n_fuels, pal_base) else
  colorRampPalette(RColorBrewer::brewer.pal(maxpal, pal_base))(n_fuels)
cols <- setNames(cols_vec, all_fuels) # Named vector: fuel type -> color

# --- FUNCTION: Aggregate data to grid for a single year ---
make_grid_vis_year <- function(year_chr) {

  # Filter data for the given year and trim whitespace in fuel names
  pp_y <- pp_long %>%
    filter(year == year_chr) %>%
    mutate(primary_fuel = trimws(primary_fuel))

  # Convert to sf (if not already) and transform to metric coordinates
  pts <- if (!inherits(pp_y, "sf")) st_as_sf(pp_y, wkt = "geometry", crs = 4326)
  else pp_y
  pts_m <- st_transform(pts, 3857)

  # Identify which grid cells intersect with points
  ints <- st_intersects(grid_all, pts_m, sparse = FALSE)

  # Subset grid to only cells that contain points (for speed)
  grid_sub <- grid_all[rowsSums(ints) > 0, , drop = FALSE]

  # Spatial join: assign each point to a grid cell
  pts_join <- st_join(pts_m, grid_sub, left = FALSE)

  # Aggregate generation by cell and fuel type
  cell_fuel <- pts_join %>%
    st_drop_geometry() %>%
    group_by(cell_id, primary_fuel) %>%
    summarise(gen = sum(generation_gwh, na.rm = TRUE), .groups = "drop")

  # Compute total generation per cell
  cell_totals <- cell_fuel %>%
    group_by(cell_id) %>%
    summarise(total_gen = sum(gen, na.rm = TRUE), .groups = "drop")

  # Identify dominant fuel type per cell (highest generation)
  dominant <- cell_fuel %>%
    group_by(cell_id) %>%
    slice_max(order_by = gen, n = 1, with_ties = FALSE) %>%
    select(cell_id, primary_fuel)

  # Join totals and dominant fuel back to grid geometry
  grid_vis <- grid_sub %>%
    left_join(cell_totals, by = "cell_id") %>%
    left_join(dominant, by = "cell_id") %>%
    mutate(primary_fuel = replace_na(primary_fuel, "None"),
           total_gen = replace_na(total_gen, 0)) %>%
    st_transform(4326) # Back to geographic coordinates for plotting
}
```

```

# Drop empty cells and assign colors based on dominant fuel
grid_vis2 <- grid_vis %>%
  filter(!is.na(total_gen) & total_gen > 0) %>%
  mutate(primary_fuel = factor(primary_fuel, levels = sort(unique(
    grid_vis$primary_fuel))),
  hex = cols[as.character(primary_fuel)]) %>%
  mutate(hex = ifelse(primary_fuel == "None" | is.na(primary_fuel), NA_character_,
    hex))

# Add HTML tooltip text for interactive map
grid_vis2$tooltip_info <- paste0(
  "<b>Primary Fuel:</b> ", grid_vis2$primary_fuel, "<br/>",
  "<b>Generated GWh:</b> ", grid_vis2$total_gen)

# Return processed grid for the year
grid_vis2
}

# --- MAP CENTER (for initializing interactive map) ---
map_center <- c(-100, 25) # Approximate geographic center of contiguous US
|
# --- PRECOMPUTE GRIDS FOR ALL YEARS (so slider updates are instantaneous) ---
precomp <- setNames(vector("list", length(years)), years)
for(i in seq_along(years)) {
  yr <- years[i]
  message("Precomputing year ", yr)
  precomp[[yr]] <- make_grid_vis_year(yr)
}

```

8.2 Shiny app.R development

```

# PARAMETERS
token <- "pk.eyJ1Ijoiaiy1hbmd1bCIsImEioiIjBwdsbTYwbHYxN2VyMmtwcWN1cwptbHg2In0.aaBmhs0l7hEHYua4nBkJzQ"
mapdeck::set_token(token)

years <- sort(unique(results_wide$year))
map_center <- c(-100, 37)

ui <- fluidPage(
  titlePanel("USA Power Plant Energy Production in Gigawatt-Hours (2013-2019)",
  sidebarLayout(
    sidebarPanel(
      # ♦ Single YEAR slider (animates automatically)
      sliderInput(
        "year",
        "Year",
        min = min(years),
        max = max(years),
        value = min(years),
        step = 1,
        sep = "",
        animate = animationOptions(interval = 1000, loop = TRUE)
      ),
      selectInput("keep_fuel", "Filter by Fuel Source Type",
        choices = c("All", all_fuels), selected = "All"),
      checkboxInput("show_legend", "Show Legend", value = TRUE)
    ),
    mainPanel(
      mapdeckOutput("map", height = "600px"),
      uiOutput("legend_ui"),
      tags$hr(),
      plotOutput("linePlot", height = "250px")
    )
  )
)

server <- function(input, output, session) {

```

```

# --- INITIAL MAP ---
output$map <- renderMapdeck({
  mapdeck(token = token, style = mapdeck_style("dark"),
    location = map_center, zoom = 4, pitch = 60)
})

# --- UPDATE MAP AND LINE CHART TOGETHER ---
observeEvent(c(input$year, input$keep_fuel, input$show_legend), {

  yr <- as.character(input$year)
  grid_vis2 <- precomp[[yr]]

  # Fuel filter
  if (input$keep_fuel != "All") {
    grid_vis2 <- grid_vis2 %>% filter(as.character(primary_fuel) ==
input$keep_fuel)
  }

  # Update map
  mapdeck_update(map_id = "map") %>%
    clear_polygon(layer_id = "grid_cells") %>%
    add_polygon(
      data = grid_vis2,
      layer_id = "grid_cells",
      fill_colour = "hex",
      elevation = "total_gen",
      elevation_scale = elev_scale,
      palette = NULL,
      auto_highlight = TRUE,
      tooltip = "tooltip_info",
      update_view = FALSE
    )

  # Legend UI
  output$legend_ui <- renderUI({
    if (!input$show_legend) return(NULL)
    fuels_here <- sort(unique(as.character(grid_vis2$primary_fuel)))
    tags$div(style = "display:flex; flex-wrap:wrap; gap:8px; margin-top:8px;",
      lapply(fuels_here, function(f) {
        tags$div(style = "display:flex; align-items:center; margin-right
:8px;",
          tags$div(style = paste0(
            "width:18px; height:12px; background:", cols[f],
            "; margin-right:6px; border:1px solid #444;"
          )),
          tags$span(f, style = "font-size:12px;")
        )
      })
    )
  })
})

```

```

# --- LINE CHART (always starts from 2013 up to current year) ---
output$linePlot <- renderPlot({

  fuel_cols <- if (input$keep_fuel == "All") all_fuels else input$keep_fuel

  data_long <- results_wide %>%
    select(year, all_of(fuel_cols)) %>%
    pivot_longer(cols = -year, names_to = "fuel", values_to = "Gwh") %>%
    filter(year >= min(years), year <= input$year)

  ggplot(data_long, aes(x = year, y = Gwh, color = fuel)) +
    geom_line(size = 1.5) +
    geom_point(size = 3) +
    scale_color_manual(values = cols) +
    scale_x_continuous(
      breaks = years,
      limits = range(as.numeric(years))
    ) +
    labs(
      title = paste0("Generation Trends by Fuel Type (2013-", input$year, ")"),
      x = "Year",
      y = "Total Generation (Gwh)"
    ) +
    theme_minimal() +
    theme(legend.position = "none")
})

}, ignoreNULL = FALSE)
}

shinyApp(ui, server)

```